

SQL Data Warehouse – View Descriptions

Documentation for the customer dimension, product dimension, and sales fact views used in a star-schema data warehouse.

1. dim_customers.sql

Description

The **dim_customers** view is a customer dimension that consolidates customer information from multiple CRM and ERP source tables into a single, analytics-ready dataset.

It combines customer identity, personal information, location, demographic attributes, and account creation data. It also resolves inconsistencies between gender fields from different source systems.

Columns

Column	Description
customer_key	A generated surrogate key that uniquely identifies each customer within the data warehouse.
customer_id	The original customer ID from the CRM customer table.
customer_number	The customer's business/customer reference number used to identify the customer in the source system.
first_name	Customer's first name.
last_name	Customer's last name.
country	The country associated with the customer's location.
marital_status	Customer's marital status.
gender	Customer's gender. Conditional logic is used to resolve inconsistencies between CRM and ERP values. Missing values are represented as n/a.
birthday	Customer's date of birth.
create_date	Date when the customer record was created in the CRM system.

Data Sources

silver.crm_cust_info
silver.erp_CUST_AZ12
silver.erp_LOC_A101

Purpose

Customer analysis, segmentation, demographic analysis, and joining customer information with transactional sales data.

2. dim_products.sql

Description

The **dim_products** view combines product information from CRM and ERP sources into a clean, analytics-ready product dimension.

It contains product identifiers, names, categories, subcategories, maintenance information, costs, product lines, and product lifecycle dates. Only currently active products are included.

Columns

Column	Description
product_key	A generated surrogate key that uniquely identifies each product within the data warehouse.

Column	Description
product_id	The original product ID from the CRM product table.
product_number	The standardized product reference number. Underscores in the original product key are replaced with hyphens for consistency with sales data.
product_name	Name of the product.
category_id	Identifier of the product category.
category	Main category to which the product belongs.
subcategory	More specific classification of the product within its category.
maintenance	Maintenance-related information associated with the product.
cost	Product cost recorded in the source system.
product_line	Product line or business segment associated with the product.
start_date	Date from which the product became active or valid.

Data Sources

silver.crm_prd_info
silver.erp_PX_VAT_G1V2

Purpose

Product analysis, category-level reporting, cost analysis, and joining product attributes with sales transactions.

3. fact_sales.sql

Description

The **fact_sales** view is the central sales transaction fact table of the data warehouse.

It combines sales transactions with customer and product dimensions, allowing transactional sales data to be analyzed together with descriptive customer and product attributes. Each row represents a sales transaction/order line.

Columns

Column	Description
order_number	Unique sales order number associated with the transaction.
product_key	Surrogate key linking the transaction to the dim_products product dimension.
customer_key	Surrogate key linking the transaction to the dim_customers customer dimension.
order_date	Date on which the sales order was placed.
shipping_date	Date on which the order was shipped.
due_date	Expected or due date associated with the order.
sales_amount	Total sales amount generated by the transaction.
quantity	Number of units sold in the transaction.
price	Unit selling price of the product.

Data Sources

silver.crm_sales_details
gold.dim_products
gold.dim_customers

Purpose

Sales performance, sales by product/customer/category, sales volume, average selling price, purchasing behavior, product performance, and order/shipping analysis.

4. Repository Description

Sales Data Warehouse – SQL Views

This project contains SQL views designed as part of a data warehouse and analytics layer. The views transform and integrate CRM and ERP data into a structured **star schema** consisting of customer and product dimensions and a central sales fact table.

Project Structure

dim_customers.sql – Creates the customer dimension by combining CRM customer data with ERP demographic and location information.

dim_products.sql – Creates the product dimension by integrating CRM product information with ERP product category data. Only currently active products are included.

fact_sales.sql – Creates the sales fact table by combining sales transactions with the customer and product dimensions.

Data Warehouse Model

The three views form a simple star schema. **fact_sales** serves as the central transactional fact table, while **dim_customers** and **dim_products** provide descriptive attributes for analytical reporting.

Data Flow

CRM / ERP Source Tables → Silver Layer → Transformation & Data Integration → Gold Layer → Customers / Products / Sales

Purpose

The purpose of this project is to transform raw CRM and ERP data into structured, business-ready datasets for analytical reporting and business intelligence. The resulting model enables analysis of sales performance across customers, products, categories, and time periods while maintaining clear relationships between transactional and descriptive data.